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A Machine Learning–Based Flood Risk Assessment Framework for Diverse Flood Regimes: A Case Study of Chennai with Comparative Analysis in Uttarkashi

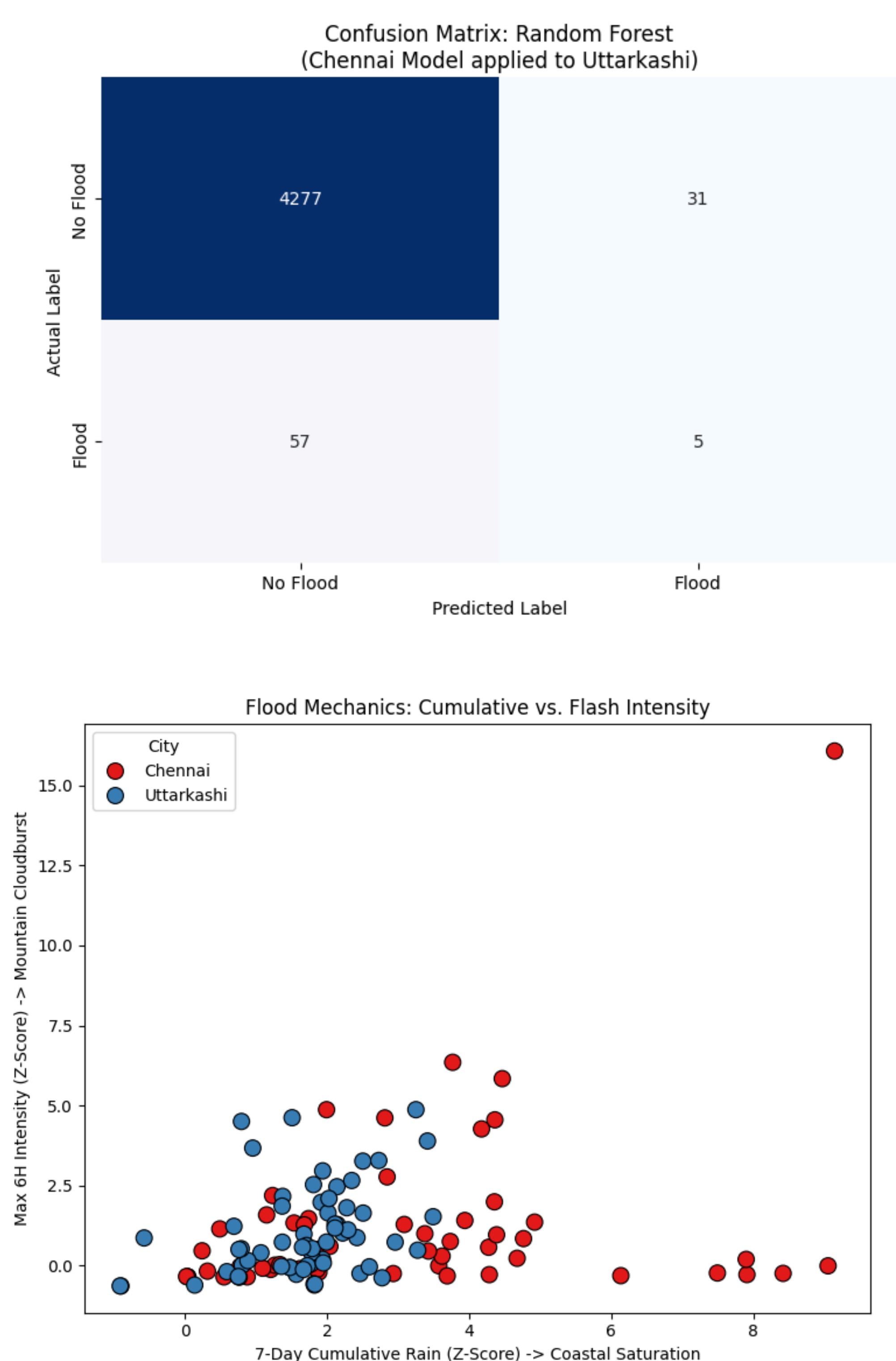
Abstract

This study develops a machine learning–based framework to assess flood risk in Chennai and evaluates its applicability in Uttarkashi. Using Logistic Regression and Random Forest models trained on historical and environmental data, the framework identifies key vulnerability factors. Results reveal significant performance degradation when applied to Uttarkashi, with 91.9% of flood events missed, highlighting the limitations of region-specific models and the need for terrain-aware approaches.

Problem Statement

- Flood risks differ significantly between Chennai (urban flooding) and Uttarkashi (terrain-driven flooding and landslides).
- Existing models are region-specific and fail to generalize across different geographic conditions.
- There is a need for a unified ML framework that captures both cumulative and intensity-driven flood mechanisms.

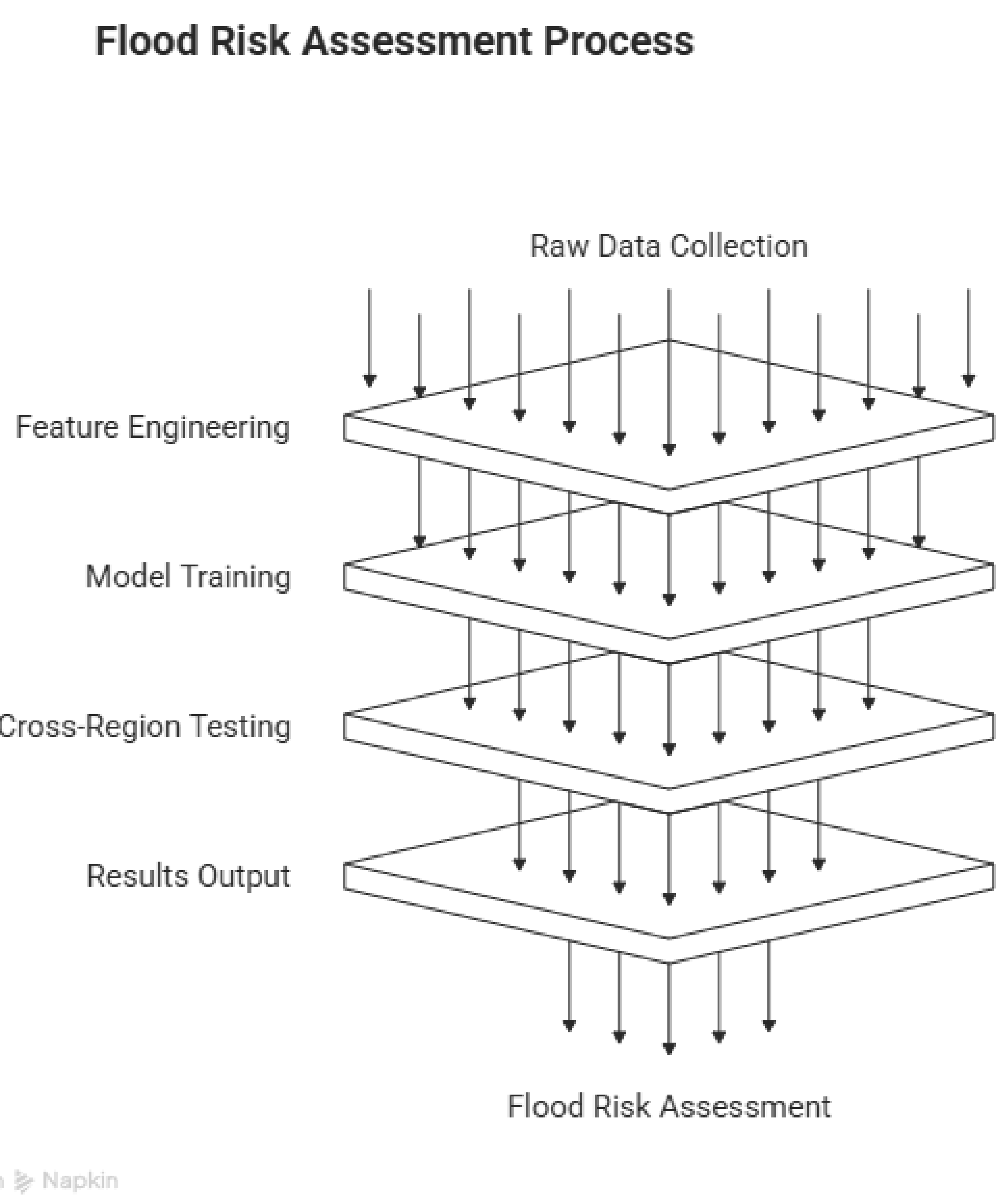
Results



The Chennai-trained model fails in Uttarkashi, missing 91.9% of flood events, highlighting poor cross-region generalization.

This performance drop is attributed to fundamental differences in flood mechanisms between coastal and mountainous regions.

Architecture & Methodology



Methodology:

- Rainfall data + feature engineering
- Models: Logistic Regression, Random Forest
- Trained on Chennai, tested on Uttarkashi

Conclusion

The framework accurately predicts flood risk in Chennai but fails in Uttarkashi due to differing flood mechanisms. Coastal flooding is driven by cumulative rainfall, while mountainous regions depend on short-duration intensity. This highlights the need for terrain-aware models and the limits of transferring ML models across diverse geographies.

References

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2. A. Arpita et al., "Integrated flood risk-based land use mapping using machine learning and Geospatial technologies: a case study of Baleswar CD block, India," Discover Cities, vol. 2, no. 1, 2025.